

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)



PRESENTS



INNOVATION | DEVELOPMENT | EXECUTION | ADVANCEMENT

Ai - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		NamoFans	
PROBLEM STATEMENT TITLE		PS3 Tracking of Funds within Bank for Fraud Detection	
	NAME	GENDER	AREA OF EXPERTISE
Member 1	Animesh Raj	Male	ML/AI & Graph Neural Networks
Member 2	Devansh Gupta	Male	Backend Engineering & System Design
Member 3	Prem Agarwal	Male	Full-Stack & Data Visualization
Member 4	MD. Faizan Khan	Male	NLP, LLMs & Compliance

IDEA TITLE

TRACE.ai – Transaction Risk Analysis & Compliance Engine

Graph-Powered Fund Flow Intelligence for Anti-Money Laundering

THE PROBLEM

₹71,543 Cr Lost to fraud in FY 24-25 (RBI).

Rule-based systems have 95%+ false positives, missing complex patterns like:

- Multi-hop layering
- Round-tripping
- Account structuring

Manual STR tracing takes **4-6 Hours**.

THE SOLUTION: DYNAMIC TEMPORAL GRAPHS

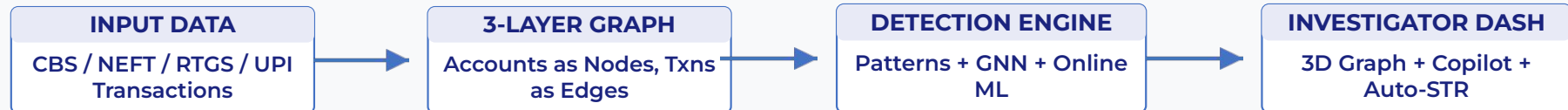
1. Graph Pattern Matcher: NetworkX algorithms for money mule fan-in/out & circular flows.

2. Temporal GNN: ChronoWave-GNN classifies suspicious nodes from evolving structural patterns.

3. Online Anomaly Scorer: HalfSpaceTrees build per-account baselines via streaming learning.

4. LLM Copilot: Auto-generates FIU-IND compliant STR packages in <5 minutes.

TRACE.ai ARCHITECTURE WORKFLOW



OUTLINE OF SOLUTION



Temporal GNN with Edge-Centric Modeling

Unlike static GNNs, uses Temporal Graph Networks (TGN) with edge-centric encoder to capture evolving fraud ring formation and layering bursts over time. Multi-head attention fusion (THG-OAFN) combines structural & temporal features, achieving AUC > 96%.



Online ML - Zero Batch Retraining

Streaming online learning using River HalfSpaceTrees builds individual behavioral baselines with ultra-low latency (<1ms). ADWIN drift detection automatically adapts to evolving fraud patterns, eliminating staleness.



Natural Language Investigation Copilot

Translates plain English questions into graph queries, executing them to return instant visual highlights and risk scores. No Cypher, SQL, or technical skills required for complex investigations.



Automated FIU-IND STR Generation

One-click generation of regulatory-compliant Suspicious Transaction Reports (STRs). Compiles fund trails, risk scores, AI analysis, and recommended actions into a structured A4 PDF in minutes (4-6 hours → 5 minutes).



Hot-Reloadable Compliance Rule Engine

YAML-defined regulatory rules (CTR, KYC, RBI Directions) update in real-time without code changes or system restarts. New compliance rules take effect immediately with first-match-wins evaluation.

TECHNICAL APPROACH

TRACE.ai ARCHITECTURE

LAYER 0: DATA INGESTION

CBS Feeds · NEFT/RTGS/UPI/IMPS · IBM AMLSim
→ (sender, receiver, amount, timestamp, channel)

LAYER 1: GRAPH ENGINE

NetworkX MultiDiGraph | Neo4j (production)
Nodes = Accounts · Edges = Transactions

LAYER 2: DETECTION ENGINE

A: Pattern Matcher (NetworkX)
5 AML typologies

**B: Temporal GNN (PyG TGN
+ ChronoWave + THG-CAFN)**

C: Online Anomaly Scorer
(River HST + ADWIN drift)

D: Compliance Rule Engine (YAML, hot-reload)

E: Risk Fusion (weighted composite score)
 $\text{pattern}(0.30) + \text{GNN}(0.30) + \text{anomaly}(0.20) + \text{compliance}(0.20) \rightarrow \text{LOW} / \text{MEDIUM} / \text{HIGH} / \text{CRIT}$

LAYER 3: INTELLIGENCE

F: LLM Alert Explainer
G: NL Investigation Copilot (NL → Graph Query)
H: Auto-STR Generator (FIU-IND compliant PDF)

LAYER 4: DELIVERY

FastAPI (REST + WebSocket)
React: 3D Force Graph · Alert Stream · Copilot Chat
KPI Dashboard · One-click STR Download

TECH STACK

Layer/Component	Technology
- Data	IBM AMLSim · Pandas · Faker
- Graph Engine	NetworkX · Neo4j Community + GDS
- GNN	PyTorch Geometric – TGN + ChronoWave-style encoder
- Imbalance	GraphSMOTE · Focal Loss
- Online ML	River – HalfSpaceTrees · ADWIN
- Compliance	YAML rule engine (hot-reloadable)
- LLM	GPT-4o-mini / Claude Haiku via LiteLLM
- NL → Query	Vanna.ai-inspired NL → Cypher/NetworkX
- Reporting	ReportLab (FIU-IND STR PDFs)
- Backend	FastAPI · WebSocket · SQLite
- Frontend	React · react-force-graph-3d (Three.js)
- Deploy	Docker Compose

Methodology note: Modular layered architecture, each detection module (A-E) operates independently with a standardized interface. Evaluation uses AMLSim-generated ground truth with precision/recall at multiple risk thresholds

FEASIBILITY & VIABILITY

FEASIBILITY ANALYSIS



Proven Open-Source Stack

NetworkX, PyTorch Geometric, River, FastAPI, Neo4j Community. No proprietary dependencies.



No Real Bank Data Required

IBM AMLSim generates realistic synthetic transactions with 6+ labeled AML typologies.



CPU-Free Deployment

TGN inference runs on CPU. Online anomaly scoring (River) processes in < 1ms.



Clean Architecture Mapping

4-member team aligns with ML/GNN, Backend, Full-stack, & NLP/LLM roles.

Challenges & Mitigations

Challenge	Risk	Mitigation
• No access to real CBS data	High	IBM AMLSim with Indian banking parameters, realistic labeled data
• GNN training on imbalanced data (fraud < 1%)	Medium	GraphSMOTE + focal loss + stratified temporal batching
• Rule-based baseline: 95%+ false positives	Medium	Multi-signal fusion (3 independent modules) reduces FP
• Scale, millions of transactions/day	Medium	NetworkX (prototype) → Neo4j GDS (production); online ML has no batch cycle
• Regulatory changes, frequent RBI circulars	Low	YAML hot-reload, rules update without code changes
• LLM hallucination in STR narratives	Low	Template-constrained generation; compliance officer reviews before filing



Deployment Roadmap

Phase 1 (Hackathon):

Full platform on synthetic data, all 5 detection modules + copilot + STR generation

Phase 2 (Pilot):

API integration with Union Bank CBS feeds; Neo4j for production graph storage

Phase 3 (Production):

Multi-branch deployment; federated model updates; real-time WebSocket alerting



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DEPARTMENT OF
FINANCIAL SERVICES



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यूनियन बैंक
ऑफ़ इंडिया
भारत सरकार का उपक्रम



Union Bank
of India
A Government of India Undertaking



IMPACT & BENEFITS

QUANTIFIED IMPACT

₹71,543 Cr

Bank fraud losses in India - RBI Annual Report FY 2024-25

95%+

False positive rate in rule-based AML systems - causing investigator fatigue and missed true positives

4-6 hrs → 5 min

STR preparation time reduction, with TRACE.ai auto-generated FIU-IND evidence packages

277 days → Real-time

Fraud identification time (IBM 2024 benchmark)

37% reduction

False positives: temporal GNN vs. static baselines
(Real-time TGNN for Payments, 2025)

>96% AUC

THG-OAFN temporal-aware models on fraud detection benchmarks - our architectural inspiration

STAKEHOLDER BENEFITS

For Union Bank of India

- Reduced RBI penalty risk (up to ₹1 Cr per AML violation)
- Lower cost per investigation - fewer manual hours per STR
- Proactive detection of layering and round-tripping invisible to current systems
- First PSB with graph-native, AI-powered AML

For Compliance Officers / MLRO

- AI-explained alerts - no black-box flags
- NL investigation interface - no Cypher or SQL skills needed
- One-click STR generation - regulatory reporting becomes trivial
- Reduced alert fatigue - focus on genuine threats

For the Financial System

- Stronger defense against money laundering and terror financing
- Better FATF Mutual Evaluation compliance for India
- FIU-IND receives higher-quality STRs with complete AI-enriched fund trails

BUSINESS MODEL

BUSINESS MODEL OVERVIEW

B2B SaaS, annual subscription licensing per bank. TRACE.ai replaces manual rule-based AML with AI-powered graph intelligence. Banks pay for reduced regulatory risk, lower investigation costs, and automated compliance reporting, all mandatory, non-discretionary expenditures.

Tier	Target Banks	Annual License
Enterprise	Top 12 PSBs + top 10 private banks	₹1–2 Cr/year
Mid-Tier	Remaining private banks	₹50L–1 Cr/year
SMB	RRBs (43), cooperative banks	₹10–30L/year

ADDITIONAL REVENUE STREAMS

Implementation consulting (one-time) · Custom typology model training · Annual compliance rule pack updates

COMMERCIALIZATION & SCALABILITY

- TAM:** Indian RegTech market ₹2,000+ Cr (25% CAGR); Global AML \$5.8B by 2028
- SAM:** ~86 banks at avg ₹75L/year = ₹64.5 Cr
- SOM Year 1–2:** Union Bank pilot → 3–5 PSBs = ₹2–5 Cr (Y1) / ₹10–15 Cr (Y2)

GO-TO-MARKET

- 1 Union Bank pilot, validate on real CBS data, measure FP reduction
- 2 IBA/DFS endorsement → 3–5 PSBs (hackathon lineage as distribution moat)
- 3 Private bank expansion + NBFC market
- 4 International: SE Asia, Middle East (FATF-aligned regulatory modules)

COMPETITIVE ADVANTAGE

Factor	Incumbents	TRACE.ai
Cost	₹5–20 Cr + multi-year implementation	₹50L–2 Cr, deploys in weeks
Technology	Rules-based + basic ML	Temporal GNN + Online ML + LLM
False Positives	95%+	Target <15%
India-specific	Heavy customization	Built for RBI/FIU-IND natively
Investigation	Manual query building	NL copilot, ask in English

RESEARCH & REFERENCES

RESEARCH PAPERS

1. **ChronoWave-GNN** (2026): Edge-centric temporal GNN with multi-scale wavelet features; SOTA on Elliptic, Ethereum, AMLSim. Direct inspiration for Module B.
2. **THG-OAFN**, Wei & Lee, PLOS ONE (2025): Uses GraphSMOTE + attention fusion; achieves AUC 96.56%, recall 95%+. Inspiration for class imbalance handling.
3. **TGN**, Rossi et al., arXiv:2006.10637 (2020): Foundation temporal graph network featuring temporal memory + message passing on dynamic graphs.
4. **GNN Fraud Detection Survey**, arXiv (2024): Reviews 100+ studies, finding GNNs consistently outperform traditional methods on financial networks.
5. **Elliptic2 Dataset** (2024): Contains 122K labeled laundering subgraphs; validates Module A typologies (layering, round-tripping, fan-in/out).
6. **BIS LVPS Anomaly Framework** (2023): Two-layer supervised + unsupervised detection; >90% accuracy on large-value payment systems. Validates Module C.
7. **Real-time TGN for Payments** (2025): Achieves 37% FP reduction + >98% accuracy. Source of key impact metric.
8. **Typology Detection via Graph Analytics**, Journal of Anti-Corruption (2025): Demonstrates ~80% typology accuracy; validates graph-metric pattern detection.
9. **Weber et al., AML in Bitcoin**, arXiv:1908.02591 (2019): Elliptic dataset GNN benchmark; foundational AML + GNN reference.

Open-Source Building Blocks:

1. **IBM/AMLSim**: Synthetic AML transaction data generator
2. **pyg-team/pytorch_geometric**: GNN framework with TGN implementation
3. **online-ml/river**: Online ML, including HalfSpaceTrees + ADWIN
4. **wei4zheng/THG-OAFN**: Temporal-heterogeneous graph attention + GraphSMOTE
5. **vasturiano/react-force-graph**: 3D force-directed graph visualization

Regulatory References:

- ❖ RBI Master Direction on KYC (2016, updated 2024)
- ❖ Prevention of Money-Laundering Act (PMLA) 2002 + PMLA Rules 2005
- ❖ FIU-IND STR Filing Manual & Guidelines
- ❖ FATF Mutual Evaluation Recommendations for India

SUBMISSION LINK

One-page summary of TRACE.ai idea (Mandatory PDF Submission):
https://drive.google.com/file/d/1c8Avh8pXRxwT3u_5EX5m31UiW8CHwdNu/view?usp=sharing